Data Analysis and Tax Compliance

Block 2A: Statistical inference

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Today: from samples to decisions

Open the interactive activity

- Define a population, parameter, sample and estimator.
- Distinguish sampling variation, selection bias and precision.
- Build and interpret confidence intervals and hypothesis tests.
- Communicate effect size, uncertainty and limitations.

Discussion

4 November, 14:30–16:30. We work conceptually today; R has a dedicated lesson.

Start with the question, not the test

A tax administration wants to estimate the proportion of filings with a discrepancy in a specified register and year.

Define the target

One filing per unit; a clearly defined reference date; a consistently measured discrepancy indicator.

A detected discrepancy is not, by itself, proof of intentional evasion.

Discussion

Would the proportion found in targeted audits answer this population question?

Population and sample; parameter and statistic

	Population	Sample
Size	N	n
Mean amount	μ	$\bar{X}$ / observed $\bar{x}$
Discrepancy proportion	π	$\hat{P}$ / observed $\hat{p}$

An estimator is a rule applied to a random sample. An estimate is its realised numerical value.

Discussion

Which quantities would change if we drew a different sample?

How records enter the sample matters

- Simple random sampling without replacement: every size-n subset is equally likely.
- Other probability designs can use unequal but known inclusion probabilities.
- Targeted audits prioritise selected cases; their unweighted yield need not estimate the population rate.
- Coverage, nonresponse and measurement error can remain even with large n.

A register where the truth is known

Our teaching register contains 1,000 **synthetic** filings.

- Mean declared amount: 93.678 thousand EUR.
- Discrepancy proportion: 19%.
- Largest-amount quarter: 250 filings.

In that priority group:

- Mean amount: 205.072 thousand EUR.
- Discrepancy proportion: 38%.

These are constructed examples, not empirical tax estimates.

Discussion

What population does a random sample from the priority group represent?

One sample is one possible outcome

E-tivity 3, section 2: 5 minutes

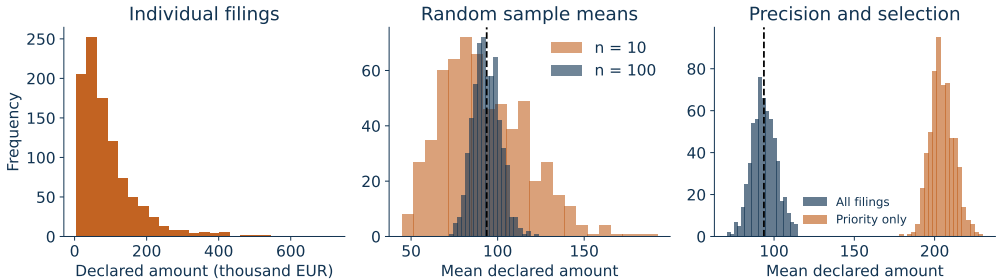
Draw two samples of 40 filings. Compare both means and both proportions. Keep the sampling rule fixed.

Sampling error is the difference between a statistic and its target parameter, for example $\bar{x} - \mu$. Different estimates do not imply different arithmetic rules: different records entered the sample.

Discussion

What would repeated samples allow us to measure?

Data distribution versus sampling distribution



Each observation in the middle and right panels is a **sample mean**. Increasing sample size concentrates estimates; restricting selection can shift their centre.

Standard deviation is not standard error

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{(i.i.d. sampling)}.$$

- SD describes differences among individual records.
- SE describes variation of an estimator across repeated samples.
- For independent sampling, quadrupling n halves the SE.

Discussion

If SE decreases, must individual declared amounts become less variable?

Without replacement: finite-population correction

For a simple random sample of n from a finite register of size N, using population variance with divisor N:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}, \quad \mathrm{SE}(\hat{P}) = \sqrt{\frac{\pi(1-\pi)}{n}}\sqrt{\frac{N-n}{N-1}}.$$

- At $n = N$, sampling uncertainty about the register total/mean is zero.
- The discrepancy count is hypergeometric, not exactly binomial.
- A census still can have coverage or measurement problems.

What makes an estimator useful?

Unbiasedness $E(\hat{\theta}) = \theta$ under the specified design/model.

Consistency $\hat{\theta} \to \theta$ in probability as information grows under suitable conditions; finite-sample bias can exist.

Efficiency Compare variance among suitable estimators of the same target under the same assumptions.

Discussion

A precise estimate can be systematically wrong for the intended target. Explain using the priority-group simulation.

The central limit idea: what becomes normal?

For i.i.d. observations with finite variance, the standardised sample mean approaches a standard normal distribution under the classical CLT.

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1).$$

- The distribution of the **mean** can become approximately normal.
- The original observations need not become normal.
- Required sample size depends on shape and tails; no universal cutoff solves selection bias.

An estimate needs a measure of uncertainty

Separate processing-time example

25 fictional independent cases; $\bar{x} = 11.4$ days; $s = 2.3805$ days. The benchmark of 10 days is illustrative, not a legal deadline.

$$\text{confidence interval} = \text{estimate} \ \pm \ \text{critical value} \times \text{SE}.$$

The target is the **mean** processing time under a stated model, not the time of one individual case.

Discussion

What should happen to the interval when variability or confidence increases?

Known σ: a z interval

$$\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Confidence	90%	95%	99%
Critical z	1.645	1.960	2.576

Exact for independent normal observations with known σ; otherwise an approximation needs justification.

Hypothetical example

$\bar{x} = 11.4$, known $\sigma = 2$, $n = 25$: 95% CI is $11.4 \pm 0.784 = [10.616, 12.184]$ days.

Unknown σ: Student's t

$$\bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

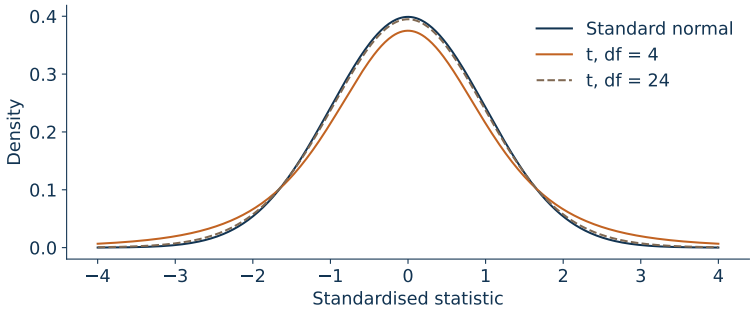
Estimating σ adds uncertainty. The t distribution has heavier tails; its critical values approach normal critical values as degrees of freedom grow.

For the supplied sample

$df = 24$, $t_{.975,24} = 2.0639$; $SE = 2.3805/5 = 0.4761$ days.

Exact small-sample inference requires independent normal observations. Strong skewness and influential values need attention.

Reading t: quantiles and tails



For a two-sided 95% interval, look up quantile 0.975 or upper-tail probability 0.025. With $n = 25$, use 24 degrees of freedom.

Worked interval: processing times

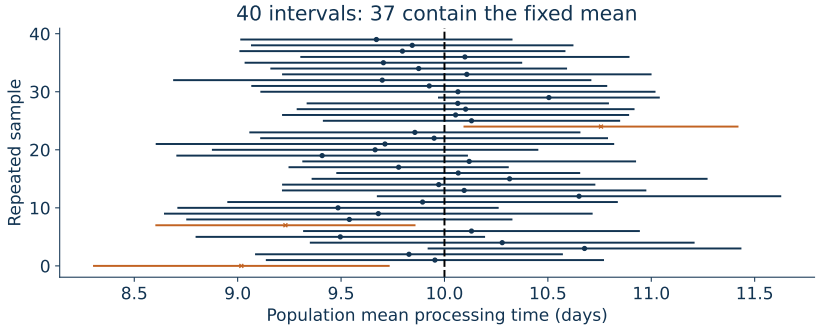
$$11.4 \pm 2.0639(0.4761) = [10.4174,\ 12.3826] \text{ days.}$$

- Point estimate: 11.4 days.
- Margin of error: 0.9826 days.
- Interval for the difference from 10 days: $[0.4174, 2.3826]$.

Discussion

Try the interval calculator. Change only one of n, SD and confidence at a time. Explain each change in width.

95% is a property of the procedure



Repeated 95% intervals cover the fixed mean about 95% of the time under the assumptions. One realised interval either contains it or does not.

What a confidence interval does not say

- It is not an interval containing 95% of individual processing times.
- It does not assign a frequentist posterior probability of 95% to the fixed mean.
- It does not correct a biased sampling frame or poor measurement.

Discussion

E-tivity 3, section 6: simulate 100 intervals. Why need observed coverage not equal the nominal percentage exactly?

Hypothesis testing: decide the comparison first

1. State H_0, the reference model, and a substantively justified H_1.
2. Choose the significance level α and one- or two-sided test.
3. Check assumptions; calculate a statistic and its tail probability.
4. Report evidence, effect size, uncertainty and limitations.

$$H_0 : \mu = 10 \qquad H_1 : \mu \neq 10.$$

Do not choose the test direction after seeing the estimate.

The one-sample t test

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{11.4 - 10}{0.4761} = 2.9406, \qquad df = 24.$$

The two-sided p-value is

$$P_{H_0}(|T_{24}| \geq 2.9406) = 0.0071.$$

Decision at $\alpha = .05$

Reject H_0. The sample provides evidence of a mean difference, under the sampling and model assumptions.

A p-value is conditional on the null model

- It is the probability of a statistic at least as extreme as observed, in the prespecified direction, assuming H_0 and the model.
- It is not the probability that H_0 is true.
- It does not measure effect size, practical importance or the chance that the entire analysis is wrong.

Discussion

If $p = .12$ at $\alpha = .05$, say "do not reject H_0", not "prove H_0".

A test for a proportion

Separate independent-trial model: 38 discrepancies in 200 cases.

$$\hat{p} = .19, \quad H_0 : \pi = .15, \quad H_1 : \pi \neq .15,$$

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}} = 1.5842, \qquad p \approx .1131.$$

Null expected counts are 30 and 170. The normal approximation is plausible here; sparse counts call for a suitable alternative.

Discussion

Why does the denominator use π_0, not $\hat{p}$?

Two possible errors, one decision rule

Reality	Do not reject	Reject
H_0 true	Correct	Type I error: α
Specific H_1 true	Type II error: β	Power: $1 - \beta$

Power depends on a **specified alternative effect**, sample size, variability, significance level and test.

Discussion

For fixed effect and variability: what changes when you increase n? What changes when you increase α?

Power and practical relevance

E-tivity 3, section 9: 7 minutes

Two-sided z test; $\sigma = 2$, $\mu_0 = 10$. Compare $n = 25$ and $n = 100$ for a true difference of 0.6 days. Then compare $\alpha = .01$ and .05.

- More information can raise power without increasing nominal α.
- Increasing α raises power but also allows more Type I errors.
- A tiny effect can be significant in a very large sample.

Statistical errors about a mean or proportion are not individual taxpayer classifications.

Write the conclusion in context

A complete comment

The estimated mean is 11.4 days (95% t CI 10.417–12.383). Against the illustrative 10-day benchmark, $t(24) = 2.941$, $p = .0071$, so we reject equality at 5%. The estimated difference is 1.4 days. This conclusion relies on the stated independent-sampling model and does not explain the cause.

Discussion

Would 1.4 days justify changing a process? What operational information is still missing?

Before the bivariate lesson

- Complete E-tivity 3 and its 12-question self-check.
- Export the short inference report and bring one unresolved question.
- Preview variable types and conditional percentages in E-tivity 4.
- The R laboratory combines sampling means and bivariate analysis.

Core lecture ends here. The following slides are optional reference material.

Reference: sample variance and degrees of freedom

$$s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n - 1}.$$

For i.i.d. data with finite variance, $E(s^2) = \sigma^2$. The signed deviations sum to zero, imposing one constraint.

- Using divisor n gives the mean squared deviation of the observed set.
- The $n - 1$ correction makes variance unbiased under the stated assumptions.
- The square root s is not generally an unbiased estimator of σ.

Reference: an interval for a proportion

The basic Wald interval is $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$, but can have poor coverage near the boundaries. The activity uses the 95% **Wilson score interval**:

$$\frac{\hat{p} + z^2/(2n) \;\pm\; z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \qquad z = 1.96.$$

For $38/200$, it is approximately $[0.1417, 0.2500]$. This independent-binomial model is distinct from sampling a sizeable fraction of a finite register.

Reference: comparing two independent means

For independent samples, Welch's statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

Welch uses approximate degrees of freedom and does not impose equal variances. A confidence interval uses the same SE.

- The pooled version assumes a common population variance.
- Before/after values on the same cases call for a paired analysis of differences.
- Repeated records are not automatically independent observations.

Reference: comparing two proportions

For independent binomial samples under $H_0 : \pi_1 = \pi_2$,

$$\hat{p}_{\text{pool}} = \frac{x_1 + x_2}{n_1 + n_2}, \qquad z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}})(1/n_1 + 1/n_2)}}.$$

- Check independence and adequate expected counts.
- A simple Wald interval for the difference uses the unpooled SE; more reliable interval methods are available.
- This is connected to the 2×2 independence test in the next block.

A small population with two values

A population contains 40 balls: 10 labelled 1 and 30 labelled 5.

$$P(X = 1) = 1/4, \qquad P(X = 5) = 3/4.$$

$$\mu = 1(1/4) + 5(3/4) = 4, \qquad \sigma^2 = 3.$$

Draw four times **with replacement**, so the draws are independent.

Discussion

The population has only two possible values. How many different values can the sample mean take?

The exact distribution of the sample mean

Let K count the draws labelled 5. Then $K \sim \mathrm{Binomial}(4, 3/4)$ and $\bar{X} = 1 + K$.

K	0	1	2	3	4
$\bar{X}$	1	2	3	4	5
$P(\bar{X})$	$1/256$	$12/256$	$54/256$	$108/256$	$81/256$

$$E(\bar{X}) = 4, \qquad \mathrm{Var}(\bar{X}) = 3/4.$$

The 16 ordered sequences of labels are **not equally likely**. Sampling without replacement would require a hypergeometric model.

R laboratory, Part A: municipal income data

- Treat the supplied municipal file as a finite teaching population.
- Draw 200 samples without replacement at each size: $n = 10, 100, 1000$.
- Compare the population distribution with the distributions of sample means.
- Relate their spread to the standard error and finite-population correction.

The unweighted mean of municipal averages describes municipalities. It is not the national average across individual pension recipients.